

Q1. Dataset : Hospital Patient Segmentation

<u>Patient ID</u>	<u>Age</u>	<u>Monthly</u>	<u>Health Risk score</u>
P ₁	28	12000	65
P ₂	55	8500	40
P ₃	42	20000	75
P ₄	30	11000	70
P ₅	60	28000	35

SSE values : 1,600 2, 350 2,220 4,200 5,190

SSE : Sum of Squared Errors.

We are using Elbow method for predicting Correct value of K.

<u>K</u>	<u>SSE</u>	<u>Reduction</u>
1	600	-
2	350	250
3	220	130
4	200	20
5	190	10

When $K = 2$ & 3 the SSE values is reduced moreWhen $K = 4$ & 5 the SSE values is reduced is Small.This variation happens after value of $K = 3$ $\therefore$ Elbow method Suggest $K = 3$

* Based on Health risk score '3' clusters are formed.

1. Low risk | Low expense.
2. moderate risk | moderate expense
3. High risk | high expense.

Impact on Healthcare Decision-Making:

- * for low risk group - preventive care & routine monitoring
- * moderate - regular followups and checks.
- * High - intensive monitoring and daily checkups.

2. Data set: E Commerce Order Analysis.

<u>Order Id</u>	<u>product Category</u>	<u>Order value</u>
Q ₁	Electronic	55000
Q ₂	clothing	25000
Q ₃	electronics	59000
Q ₄	furniture	30000
Q ₅	clothing	27000

A retailer applies Hierarchical Agglomerative Clustering (HAC) to customer orders using approximately encoded product-category information and standardized order value.

- Sol: HAC :
- ① Calculates distance b/w clusters.
 - ② Merge two closest clusters.
 - ③ Recalculates distance
 - ④ Continues until all observations become one cluster.

* O₁ + O₃ : Cluster 1 : Electronics

* O₂ + O₅ : Cluster 2 : Clothing.

* O₄ : Cluster 3 : Furniture.



* But the optimal no. of clusters cannot be determined by given data.

Step 1: Onehot Code method.

Order	electronics	clothing	furniture
O ₁	1	0	0
O ₂	0	1	0
O ₃	1	0	0
O ₄	0	0	1
O ₅	0	1	0

Step 2: Normalisation :

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

$$X_{\min} = 25000$$

$$X_{\max} = 57,000$$

Order	electronics	clothing	furniture	Normalised
O ₁	1	0	0	0.9375
O ₂	0	1	0	0
O ₃	1	0	0	1
O ₄	0	0	1	0.1560
O ₅	0	1	0	0.0625

Step 3: Euclidean distance.

Q_1 & Q_3 (Electronics)

$$d(O_1, O_3) = \sqrt{(1-1)^2 + (0-0)^2 + (0-0)^2 + (0.9375-1)^2} \\ = 0.0625$$

next

$$d(O_2, O_5) = \sqrt{(0-0.0625)^2} \\ = 0.0625$$

* These are small distances so, it will be in same cluster.

Step 4: distance matrix:

	O_1	O_2	O_3	O_4	O_5
O_1	0	1.697	0.063	1.616	1.663
O_2	1.697	0	1.732	1.423	0.063
O_3	0.063	1.732	0	1.647	1.697
O_4	1.616	1.423	1.647	0	1.417
O_5	1.663	0.063	1.697	1.417	0

Step 5: Hierarchical Agglomerative Clustering

$$D(A, B) = \min(m, y)$$

merge 1:

$$O_1 - O_3 = 0.0625$$

$$C_1 = \{O_1, O_3\}$$

$$C_2 = \left\{ \begin{array}{l} O_2, O_5 \\ O_4 \end{array} \right\}$$

C_1, O_4	C_2, O_4
$(O_1, O_3) O_4$	$(O_2, O_5) (O_4)$
$= \min(O_1, O_4) (O_3, O_4)$	$= \min(O_2, O_4) (O_5, O_4)$
$= 1.616$	$= 1.417$

So, we can merge C_2 with O_4

So, finally 2 clusters : $C_1 - O_1, O_3$
 $C_2 - O_2, O_5, O_4$

3. Dataset : Mobile App user behaviour.

User	Daily usage	Login freq	In-App purchase
U_1	120	18	2000
U_2	60	5	5000
U_3	140	22	
U_4	50		1500
U_5	130	4	6000
		20	1800

Step 1: Analyze data and make groups:

Group 1 : High usage, high login freq, low purchase
 $\{ U_1, U_3, U_5 \}$

Group 2 : low usage, low login freq, High purchase
 $\{ U_2, U_4 \}$

Step 2: Normalize the data.

usage $\rightarrow$ 50 - 140 min

loginfreq $\rightarrow$ 4 - 22

purchase $\rightarrow$ 1500 - 600

Step 3: probability based membership.

User	Engaged primary	Highvalue cluster	Interpret
U_1	0.85	0.15	Engaged
U_2	0.20	0.80	High-value
U_3	0.95	0.05	Engaged
U_4	0.10	0.90	High-value
U_5	0.90	0.10	Engaged.

$P(C_1) + P(C_2) = 1$

Step 4:

$U_1 \rightarrow$ Engaged

$U_2 \rightarrow$ High value

$U_3 \rightarrow$ Engaged

$U_4 \rightarrow$ Highvalue

$U_5 \rightarrow$ Engaged.

Step 5:

Engaged cluster - $\{U_1, U_3, U_5\}$

High value cluster - $\{U_2, U_4\}$

Step 6: In hard clustering user belongs to only one cluster.

$U_1 \rightarrow$ Engaged = 1, Highvalue = 0 $\therefore$ no indication of uncertainty.

* Normalisation is important because k-means is based on distance calculations without normalization, the feature with the largest numerical scale, such as rainfall, can dominate the clustering process. After normalization it more meaningful, reliable, and useful agricultural clusters for decision making.

5. Data Set : Employee performance Evaluation

<u>Employee</u>	<u>productivity</u>	<u>Attendance (%)</u>	<u>Skill Assessment</u>
E ₁	88	92	84
E ₂	62	75	68
E ₃	78	88	72
E ₄	55	65	58
E ₅	93	96	90

Sol: The silhouette score measures how well employees fit within their assigned clusters. Its value generally ranges from -1 to 1. A higher score indicates that employees within the same cluster are more similar to each other and more clearly separated from employees in other clusters.

4. Dataset : Agriculture Crop Monitoring.

Farm	Rainfall	soil-fertility	crop yield (tons)
F ₁	300	8	85
F ₂	500	4	70
F ₃	400	6	80
F ₄	280	9	90
F ₅	520	3	65

Rainfall $\rightarrow 280 - 520$

Soil fertility $\rightarrow 3 - 9$

Crop yield $\rightarrow 65 - 90$

* without normalization, k-means uses distance calculations. Because rainfall has much larger numerical values, it can dominate distance calculation.

Difference in Clustering Results

Before Normalization K=3

After Normalization

- * Rainfall has excessive influence.
- * Clusters may mainly reflect rainfall
- * Soil has fertility less influence
- * All features have comparable influence.
- * Clusters reflect overall farm characteristics.
- * Soil fertility contributes approximately.

model	Silhouette Score	Interpretation
K=2	0.48	moderate
K=3	0.68	Strong & well Separated
K=4	0.52	moderate

K=3 has the highest silhouette score of 0.68 ∴ 3 clusters

∴ K provides best balance between cluster Cohesion and Separation.

Possible Employee Segmentation:

- ① High performing Employees.
- ② Moderate - performing Employees
- ③ Employees needing improvement.

Conclusion:

- * K=3 model is best.
- * more accurate Employee Segmentation, Personalized training, performance evaluation and better management decision making.

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10/10/26